

AI Fundamentals in the Work Environment

What AI is, where it already sits in your tools, and how to use it well — in plain language.

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Executive Summary

The project focused on using an AI tool to assist with the task of **explaining and summarizing a study file** in a clear and structured way. A structured prompt was provided to the AI, including instructions about the required content, level of detail, and preferred format. Before submitting the file, sensitive and important personal or organizational information was **masked to protect privacy and confidentiality**. The AI-generated output was then reviewed and refined to ensure that it addressed the requirements of the task. The results showed that using a structured prompt helped produce an explanation that was **clearer, more organized, and more detailed** than the initial approach. The AI also helped present the study material in a way that made the key concepts easier to understand and review.

Overall, the use of AI improved the efficiency and quality of the task while maintaining appropriate data privacy through masking of sensitive information.

Introduction

Artificial Intelligence (AI) is becoming an increasingly useful tool in the workplace, helping employees complete daily tasks more efficiently and effectively. This course provides a basic understanding of AI, how it can be used with common workplace tools, and how to work with AI assistants through clear prompts. It also covers responsible AI use, common limitations and errors, and the importance of checking AI-generated outputs before using them. The course focuses on practical applications of AI in everyday work without requiring any coding knowledge.

Project Objectives

Project Objectives

By the end of this capstone project, you will be able to:

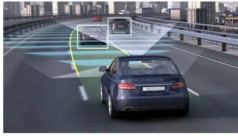
1. **Choose a suitable task.** Pick one real, repeated, text-based task from your weekly work (like drafting, summarising, or reformatting) that AI can help with and that you can check for accuracy.
2. **Write a clear, structured prompt.** Use all four parts of the R.A.C.E. framework (Role, Action, Context, Expectation) so the AI knows exactly who it should be, what to do, the background, and what the result should look like.
3. **Improve the result step by step.** Write at least one follow-up prompt that says what was wrong with the first draft (tone, format, or depth) instead of starting over.
4. **Protect private information.** Remove or replace all personal names, account IDs, and confidential company data before sending anything to the AI.
5. **Check the output and measure the benefit.** Carefully verify every line of the AI's answer for facts and numbers, then give an honest estimate of how much time you saved.
6. **Share your work professionally.** Put your final document (PDF or DOCX) on GitHub and send the link to your trainer.

Project steps

The primitive method is a direct command to artificial intelligence to explain slides by dividing the slides, copying the text, and sharing images. The result is a lot of wasted time, the need for many content divisions, and the possibility of sharing private information.

Reinforcement learning in practice is challenging

- There are also safety, privacy, and ethic concerns in exploration
 - We are dealing with unknown unknowns
 - We are learning from explorations



اشرح لي هذا النص بالعربي ثم بالانقلش بلغة بسيطة (last but not the least is in practice in reality terms there are a lot of things and you have to take into account for example the safety and privacy even ethical issues right so yeah that's I mean kind of a you can say problem area or the challenging I would say the challenging area is because there are privacy concerns okay so how (for example there they will be taken care of

- تحويل نظاماً نظائرياً يستخدم التعلم المعزز لاتخاذ قرارات الموافقة أو الرفض على طلبات القروض البنكية.
- إذا أراد هذا النظام "استكشاف" سياسة جديدة (مثلاً الموافقة على شخص كان سيُرفض عادة، لمعرفة ماذا سيحدث)، فهذا يؤثر على مشاكل أخلاقية خطيرة جداً:
 - الخصوصية: يتطلب الأمر استخدام بيانات شخصية حساسة جداً عن الأفراد (تدخلهم، تاريخهم الائتماني، إلخ).
 - الأخلاقيات والعدالة: إذا "استكشف" النظام بشكل عشوائي، فقد يرفض شخصاً يستحق القرض فعلاً (كما يظهر في الصورة: "مرفوض/Rejected" و"منكر/Denied")، أو يوافق على شخص غير مؤهل — وهذا يؤثر على حياة حقيقية لأشخاص حقيقيين، وليس مجرد تجربة بلا عواقب.

الخلاصة:

هذه الشريحة تُظهر أن التعلم المعزز، رغم قوته النظرية، يواجه عقبات عملية جدية جداً عندما نطبقه في مجالات تمس حياة البشر مباشرة — فلا يمكننا ببساطة أن "نترك" النظام يجرب ويخطئ بحرية كما يفعل في لعبة فيديو، لأن الأخطاء هنا قد تكون لها عواقب حقيقية وخطيرة وأحياناً لا رجعة فيها.

In Simple English

This slide continues discussing the **practical challenges of Reinforcement Learning**, but this time it covers **ethical and legal aspects**, not just technical ones: **Safety, Privacy, and Ethics**.

Why do these problems specifically show up during "Exploration"?

We're dealing with "Unknown Unknowns":

As we discussed earlier in the exploitation vs. exploration topic — the agent needs to **try new things it hasn't tested before** in order to learn. But the problem is, **we don't know in advance** what the outcome of this new trial will be — it could be safe, or it could be **very dangerous**.

We learn from actual exploration:

+ Write a message...

Claude is AI and can make mistakes. Please double-check responses.

Sonnet 5 Medium

The professional method started by converting the file to MD format to delete private information (masking).

Uploaded File

intro2RL.pdf
6.17 MB

Converted Markdown

Preview HTML

Copy

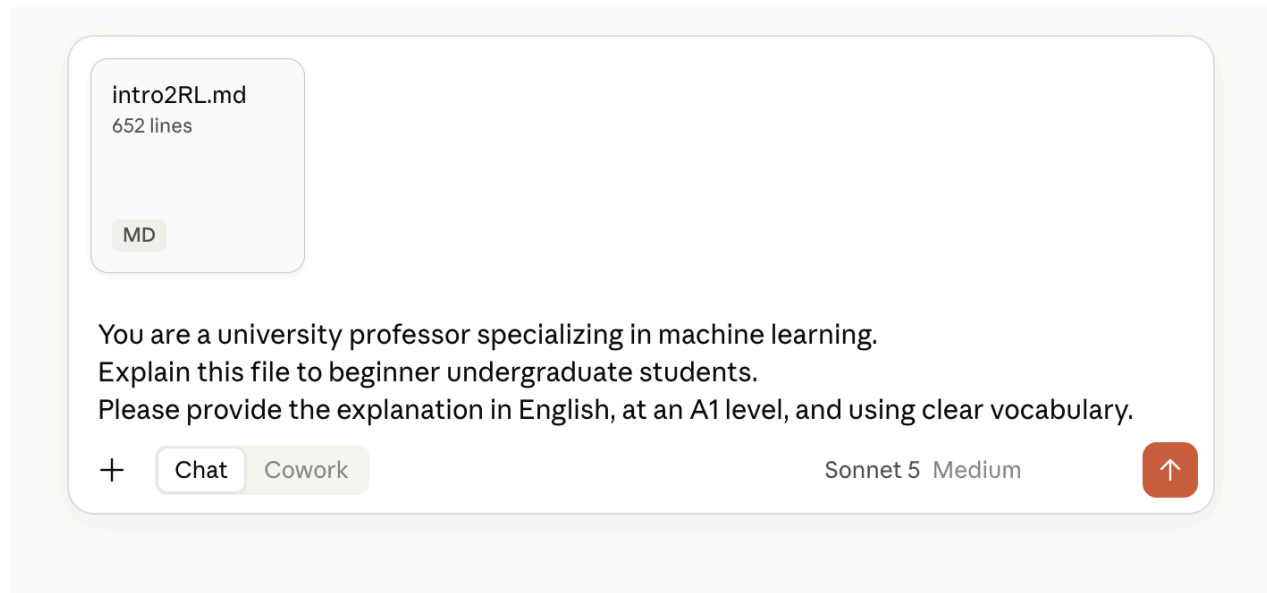
Download

```
CS@UVA 1RL2022-Fall
Outline
• What is reinforcement learning?
• Why do we need it?
• How to?
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Reinforcement learning
...

Reinforcement learning (RL) is an area of machine
...

learning concerned with how software agents ought to
take actions in an environment in order to maximize
the notion of cumulative reward.
- Wikipedia
Reinforcement learning is learning what to do — how
to map situations to actions — so as to maximize a
numerical reward signal.
- Sutton & Barto, 2018
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After deleting sensitive information from the file, I shared it with the AI using this prompt (You are a university professor specializing in machine learning. Explain this file to beginner undergraduate students. Please provide the explanation in English, at an A1 level, and using clear vocabulary.).



Conclusion and Results

The goal of this project is to apply artificial intelligence to a real, repetitive task from my weekly work: explaining a course document. I started by choosing a text task with verifiable results, then wrote a prompt using the R.A.C.E. framework with its four elements: role, action, context, and expected outcome. I then refined the first draft with a follow-up prompt that clearly identified what needed modification instead of starting over.

I also ensured privacy by removing personal names, course numbers, and course information before submitting any input. Finally, I reviewed each line of output to verify the accuracy of the information and figures before finalizing it.

Output Quality: The initial draft was close to the desired quality and required only minor adjustments. After the follow-up command, the result became clearer, more formal, and more in-depth.

Privacy: No names, numbers, or confidential information remained in the input.

Verification: I reviewed the output and found some numbers that required correction, such as a table being misread and some outputs being reversed.

Time Saved: The task previously took me significantly longer due to trial and error, revisions, and content segmentation.